KRISHNA SHARMA

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PROFILE

B.Tech. student in Mathematics & Computing at IIT Delhi with a focus on building ML systems for finance, and healthcare. Experience in building interpretable ML models and their deployment.

Year	Degree/Board	Institute	CGPA/%
2022-2026	B.Tech. in Mathematics & Computing	Indian Institute of Technology Delhi	**8.1**
2022	CBSE XII	Disha Delphi Public School, Kota	**96.8%**
2020	CBSE X	Disha Delphi Public School, Kota	**95%**

WORK EXPERIENCE

Millennium Management, Bangalore *May 2025 - July 2025*
Quantitative Developer Intern

- Built an end-to-end **anomaly detection system** for cross-asset PnL explain data, enabling faster identification of unexplained risk drivers across trading desks.
- Developed **ensemble regression models** (XGBoost, Random Forest) on **180+ financial features**, achieving $R^2 = 0.86$ and improving baseline explainability of PnL movements.
- Designed a **SHAP-based interpretability pipeline** to attribute feature-level contributions, allowing granular diagnosis of anomalies.
- Built an **LLM-based explanation layer** to convert model outputs into human-readable insights, reducing manual analysis effort.
- Deployed the system as a **Flask REST API** and integrated it into internal tool for interactive use by PMs.

TITANE, INRIA, France *May 2024 - July 2024*
Research Intern, Supervisor: Rahima Djahel

- Contributed to extending **progress monitoring and quality control** modules in a **digital twin platform** under the EU-funded Bim2Twin project.
- Implemented **mesh reconstruction pipelines** using CGAL to generate surface meshes from construction site point cloud data.
- Designed an **object matching algorithm** between BIM models and LiDAR scans using overlapping triangular mesh representations.

SCHOLASTIC ACHIEVEMENTS

- **International Junior Science Olympiad (IJSO) 2019:** Awarded **Gold Medal** as member of the **Indian national team** (6 students).
- **JEE Advanced:** Secured **AIR 276** among **155,000+** candidates; **JEE Main:** Secured **AIR 24** with **100 percentile** among **900,000+** candidates.
- **Indian National Mathematical Olympiad (INMO):** Qualified; ranked among the **top 450** students nationwide.
- **KVPY Fellow:** Secured **AIR 22 (SA)** and **AIR 140 (SX)**; received offer letter from **IISc Bangalore**.
- **International Olympiad on Astronomy & Astrophysics (IOAA):** Selected for **national camp stage** at **HBCSE, Mumbai**; ranked among **top 102**.
- **NTSE Scholar:** Scholarship awarded by the **Government of India**; ranked among **top 2000** out of **1 million candidates**.

TECHNICAL SKILLS

- **Languages:** Python, C++, Matlab
- **ML Frameworks:** PyTorch, TensorFlow, Scikit-learn, HuggingFace
- **Data & Systems:** Pandas, Flask, REST APIs

PROJECTS

Ontology-Guided Molecular Captioning *Aug 2025 - May 2026*
B.Tech. Project, under Prof. Sandeep Kumar

- Developed a **knowledge-grounded molecular captioning system** (SMILES $\rightarrow$ text) using the **ChEBI dataset (∼30K molecules)**.
- Proposed a **multi-modal architecture** combining molecular representations (MolFormer) with structured ontology embeddings (OWL2Vec).
- Introduced a **contrastive alignment objective** to jointly embed molecular, ontology, and textual representations.
- Built a **retrieval-augmented pipeline** to dynamically incorporate relevant ontology subgraphs during generation.
- Achieved **∼10% improvement over state-of-the-art baselines (e.g. MolT5)** across BLEU, ROUGE, METEOR, and BERTScore.

Multi-Agent Market Simulation for RL Trading *Nov 2025*
Reinforcement Learning Course Project, under Prof. James Arambam

- Implemented a **GAN-based multi-agent market simulator** to generate **action-conditioned financial time series**.
- Trained a **DRNN-based RL agent** in the simulated environment to overcome overfitting to static historical data.
- Reproduced **∼12% improvement in out-of-sample profit**, with improved Sharpe ratio and reduced drawdown.

Speech Emotion Recognition (SER) *Mar 2025*
Data Mining Course Assignment, under Prof. Niladri Chatterjee

- Investigated emotion extractability using unsupervised clustering by applying k-**Means** (with k equal to the number of emotion classes) on Librosa-extracted audio features (MFCCs, prosodic, chroma), achieving an **ARI of 0.673**.
- Fine-tuned **wav2vec2 / emotion2vec** transformer models for the SER task, achieving **96.5% accuracy** on the Kaggle private leaderboard and securing **first place** among 23 teams. Model weights are released on Hugging Face (**500+ downloads**).

Traffic Forecasting with DCRNN *Apr 2025*
Data Mining Course Project, under Prof. Niladri Chatterjee

- Implemented a **spatiotemporal graph neural network (DCRNN)** on METR-LA dataset.
- Achieved performance within **2% MAE of state-of-the-art**, verifying the results.

Moth Image Classification *Mar 2025*
Deep Learning Course Assignment, under Prof. James Arambam

- Implemented a configurable **ResNet (6n+2, n=2)** in PyTorch; trained on **12K images (100 classes)**, achieving $\sim 78\%$ accuracy.
- Used **Grad-CAM** for weakly supervised segmentation, achieving ~ 0.58 IoU and generating **500+ masks**.

RELEVANT COURSES

Intro to Machine Learning	Deep Learning	Reinforcement Learning	Data Mining
Computer Architecture	Operating Systems	Cryptography	Theory of Computation
Statistical Methods	Financial Mathematics	Microeconomics	Econometric Methods
Data Structures & Algos	Analysis & Design of Algos	Optimization Methods	Linear Algebra & Applications
Intro to Computer Science	Intro to Probability	Discrete Mathematics	Combinatorics